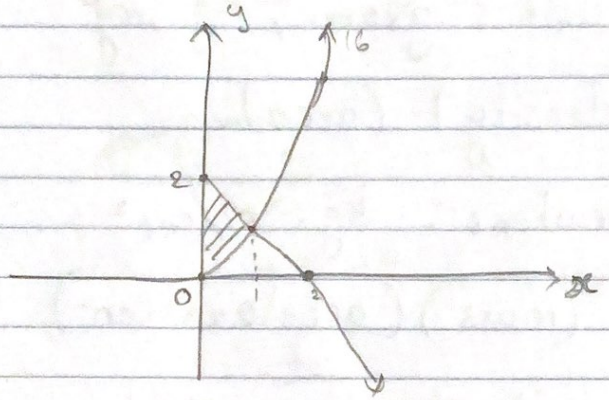
Quiz :-

C1:

$y = x^4$ and $y = 2 - |x|$



$x^4 = 2 - |x| \Rightarrow x^4 + x - 2 = 0$

$x(x^3+1) - 2 = 0 \Rightarrow x(x^3+1) = 2$, $x = 2$

$x^3 + 1 = 2 \Rightarrow x^3 = 1 \Rightarrow x = \sqrt[3]{1} = 1$

$$A = \int_0^1 \left[2 - (x) - x^4\right] dx$$

$$A = \int_0^1 (2 - x - x^4)\, dx$$

$$A = \left[2x - \frac{x^2}{2} - \frac{x^5}{5}\right]_0^1$$

$$A = \left[2 - \frac{1}{2} - \frac{1}{5}\right] - \left[0\right]$$

$$A = \frac{3}{2} - \frac{1}{5} \Rightarrow A = \frac{15 - 2}{10}$$

$$A = \frac{13}{10}$$